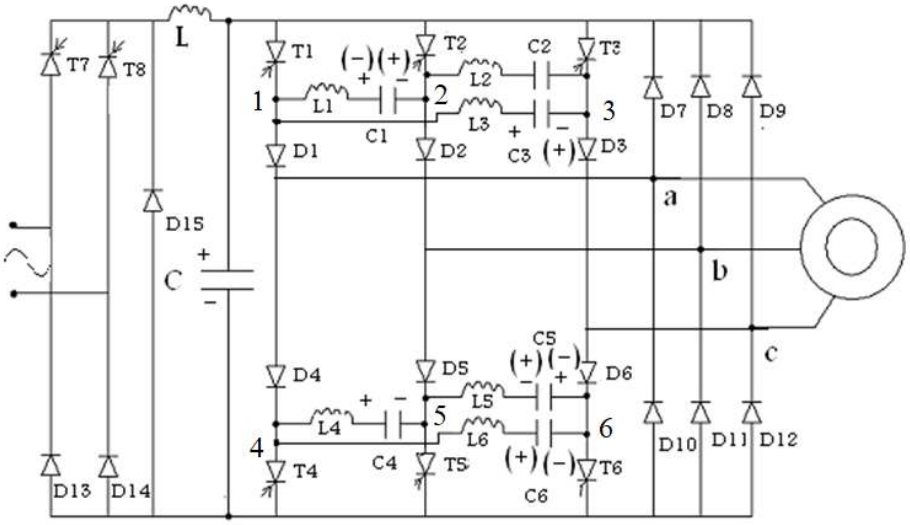
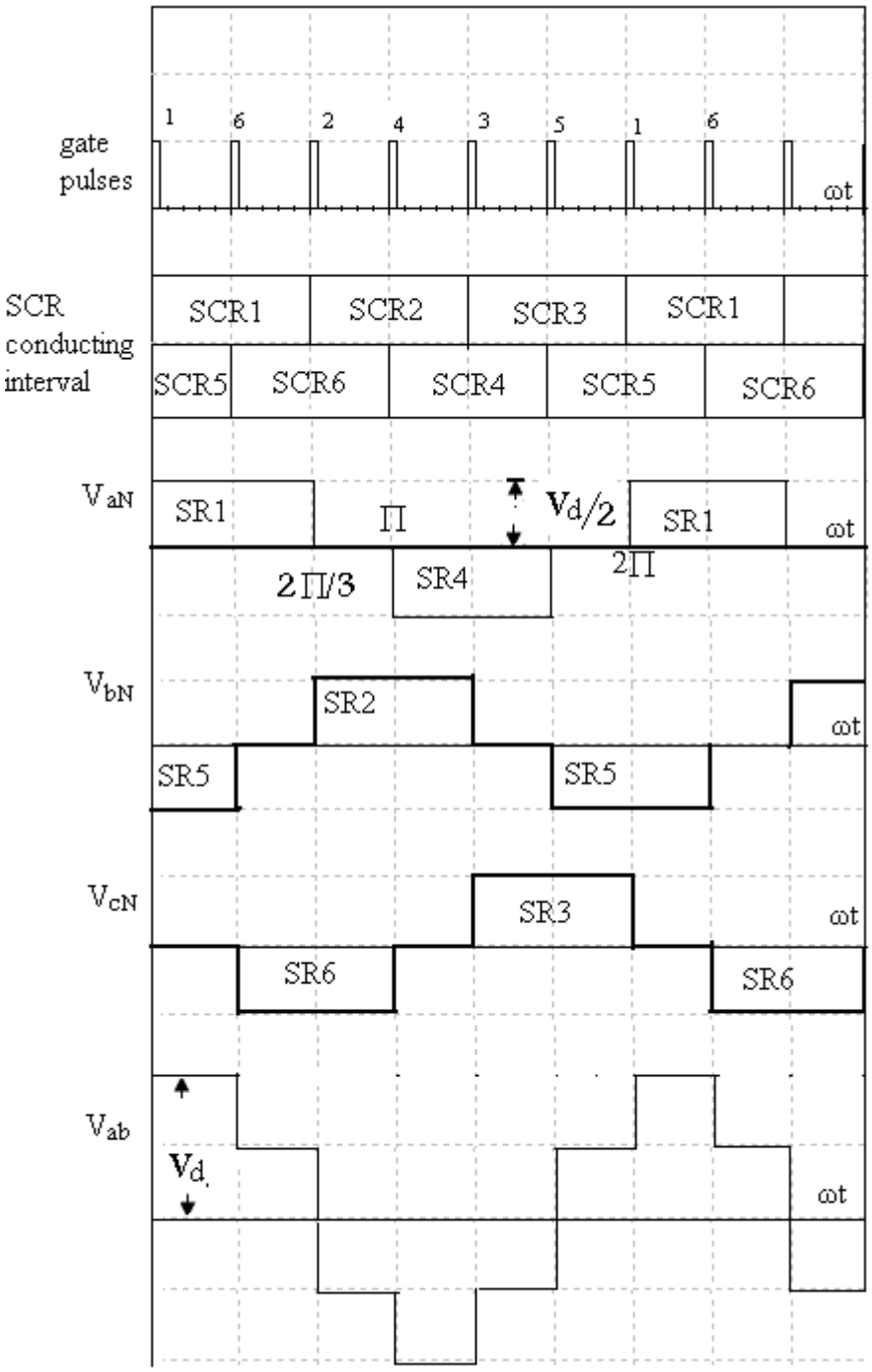
عمل مفتاحين معا. كل مفتاح يعمل 120^0

مثال :
للمحركات التي قدرتها اقل من 5 KW يمكن تقويم موجة من المصدر احادي الطور كما في الشكل بدل المصدر ثلاثي الطور.



مكونات الدائرة:
SR_8, SR_7 يتم التحكم بهما عن طريق زاوية قدح لاعطاء فولتية DC متغيرة القيمة .
D_{13},D_{14} يشكلان صمامين في جسر التقويم – كاملة التقويم.
SR_6- SR_1 ثايرستورات محكومة لتنظيم التردد عن طريق زاوية القدح.
C_1-C_6, L_1-L_6 لاحداث التبديل(commutation)المطلوب للثايرستورات (إخماد).
D_1-D_6 لمنع اتصال طورين من اطوار المحرك معا ولفصل مكثفات التغيير عن الحمل.
D_7-D_{12} لارجاع الطاقة غير الفعالة من المحرك إلى المكثف C.

مبدأ العمل:
1-على فرض أن الثايرستورات S_{R1} وS_{R5} موصولة (مغلقة) الآن يمر التيار كالتالي:
(+) ← S_{R1} ← D_1 ← الطور a ← الطور b ← D_5 ← S_{R5} ← (-) .
2-ستتشحن المكثفات C_5,C_4,C_3,C_1 كما في الشكل.
3-عند إعطاء نبضة قدح إلى الثايرستور S_{R6} ستهبط فولتية النقطة 6 إلى الصفر عندها يبدأ المكثف C5 بالتفريغ:
(+) ← S_{R6} ← S_{R5} ← L_5 ← (–) وينحاز S_{R5} عكسيا ويطفئ turned off
4-تشحن المكثفات C_6,C_5 كما في الشكل (داخل الدائرة) بينما تبقى المكثفات C_3,C_1 كما كانت شحنتها سابقا.
5 -عند إعطاء نبضة قدح إلى الثايرستور S_{R4} سيطفئ الثايرستور S_{R6} وهكذا لبقية الثايرستورات.



Three switches work at the same time and each switch operates by displaying a continuous pulse of 180 degrees:

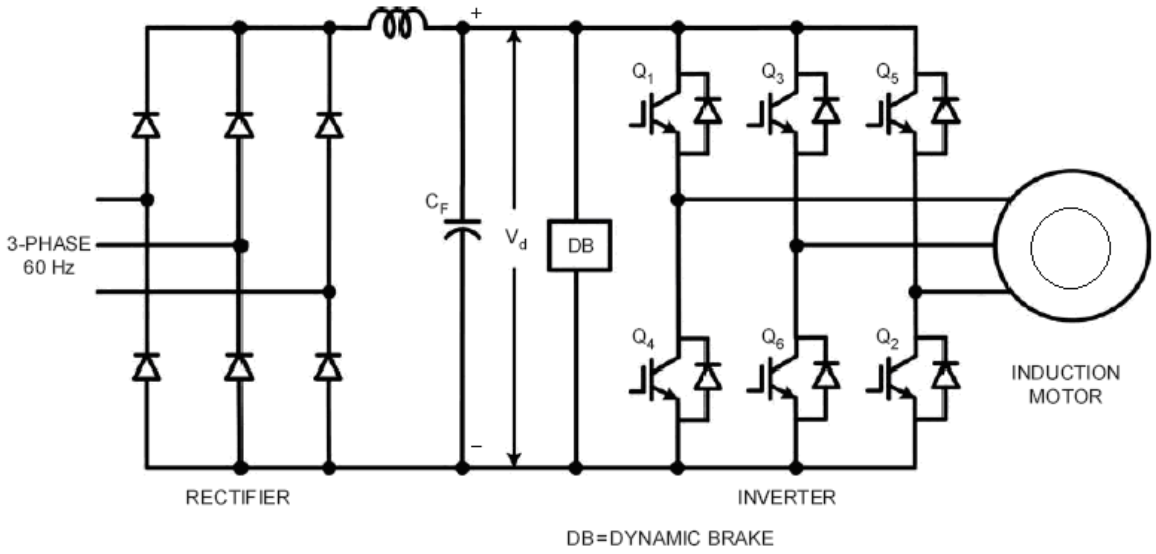


FIGURE 4.11 Three-phase bridge inverter with diode rectifier for induction motor speed control.

One of the most common converter topologies that is very widely used in industry is shown in this figure. It consists of a three-phase bridge inverter with a three-phase diode rectifier in the front end. The rectifier (which can be single or three-phase) converts ac to uncontrolled dc. The harmonics in the dc link are filtered by an *LC* or *C* filter to generate smooth voltage V_d for the inverter. The inverter consists of three half-bridges or phase legs to generate three-phase ac for industrial motor drives or other applications. For the present, neglect the element DB in the dc link. Instead of generating dc by a rectifier, it can be done with a battery, fuel cell, or photovoltaic dc source. In all such cases, V_d is usually unregulated. The battery-fed inverter drive is commonly used for electric/hybrid vehicle drives. Note that because of the diode rectifier in the front end, the converter system cannot regenerate power. The filter capacitor C_F sinks the harmonics from the rectifier as well as inverter sides.

The voltage source inverter is usually constructed with IGBT transistors and very fast parallel diodes. In the last several years, the use of IGBT modules with six transistors and six diodes has been the preferred configuration.

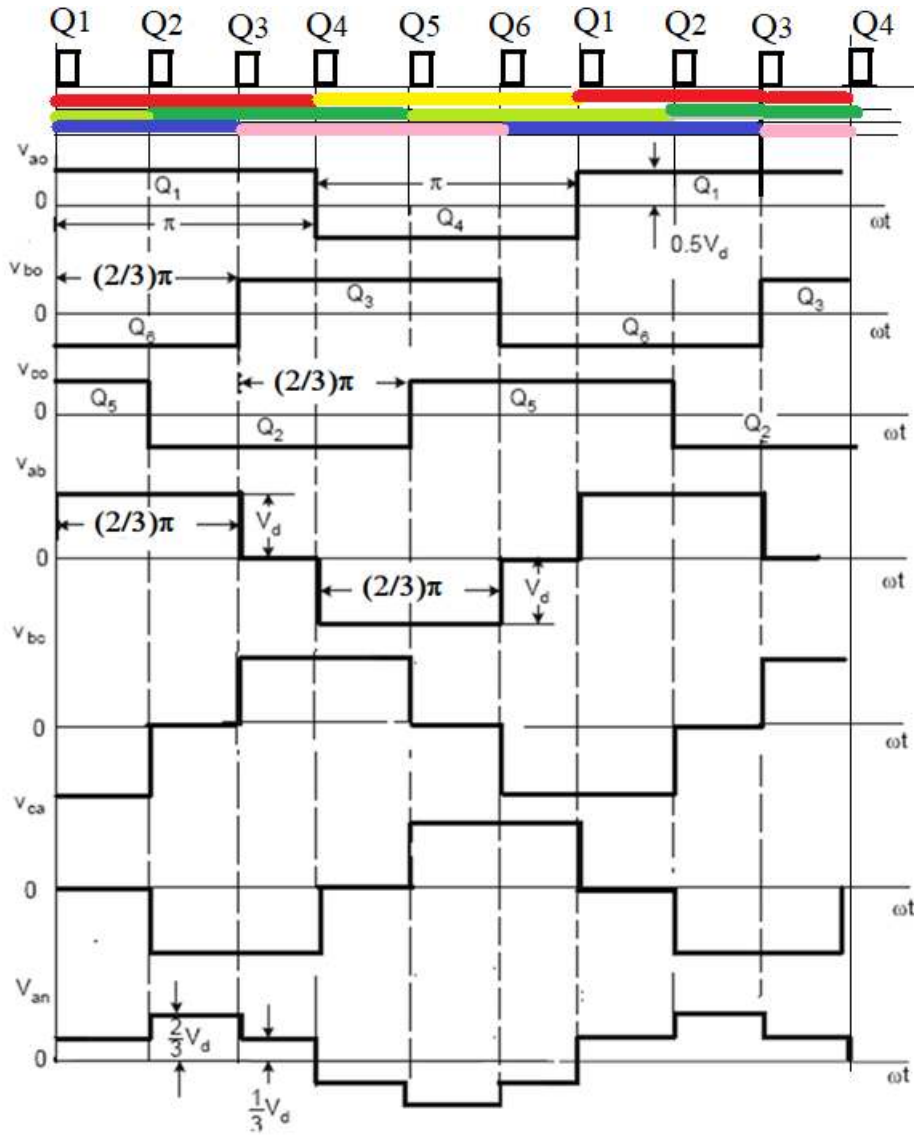


FIGURE 4.12 Three-phase bridge inverter output voltage waves in square-wave (or six-step) mode.

The three-phase inverter of Figure 4.11 can be operated in either the square-wave or PWM mode. The waveforms in square-wave mode are explained in this figure. Three phase legs of the inverter generate square waves at 120° mutual phase-shift angle, where the output phase voltage magnitudes (±0.5*Vd*) are shown with respect to the artificial dc link center point. The line voltages *vab*, *vbc*, and *vca* are constructed by subtracting the adjacent phase voltages. For an isolated neutral wye-connected load, the phase voltage wave *van*, for example, is given by the relation *van*= 2/3 *vao* – 1/3 *vbo* – 1/3 *vco* due to the absence of triplen (third or multiple of third) harmonics. The line and load phase voltages have characteristic six-step wave shapes with suppression of triplen harmonic voltages. With three-phase balanced load, the line currents are also balanced but may be rich in harmonics. In the square-wave mode, output voltage control is not possible by the inverter and *Vd* variation reflects to the output.

Comparison between 120 degrees and 180 degrees inverter control modes

NN	Parameter	120 degrees Inverter control mode	180 degrees Inverter control mode
1	Switch conduction interval	2x60 degrees	3x60 degrees
2	Switches sequence of operation	12,23,34,45,56,61	123,234,345,456,561,612
3	Output phase voltage wave form	120 degree quasi-square segments with dead zone symmetrical around time axis with 120 degrees phase shift	Six stepped segments symmetrical around time axis with 120 degrees phase shift
4	Output line voltage wave form	Six stepped segments symmetrical around time axis with 120 degrees phase shift	120 degree quasi-square segments with dead zone symmetrical around time axis with 120 degrees phase shift
5	Output voltage harmonics	Only odd harmonic present	Only odd harmonic present
6	Phase rms voltage	0.409 V_{dc}	0.47 V_{dc}
7	Line rms voltage	0.707 V_{dc}	0.816 V_{dc}
8			